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(54) SENSOR MODULE AND MEASUREMENT SYSTEM

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(57) ABSTRACT

A sensor module includes a first sensor device, a second sensor device, and a microcontroller. The microcontroller receives first detected data input from the first sensor device and second detected data input from the second sensor device. The microcontroller determines first integrated data by integrating the first detected data up to a synchronization timing of an external synchronization signal, determines second integrated data by integrating the second detected data up to the synchronization timing, and outputs the first integrated data and the second integrated data to a host device at the synchronization timing.

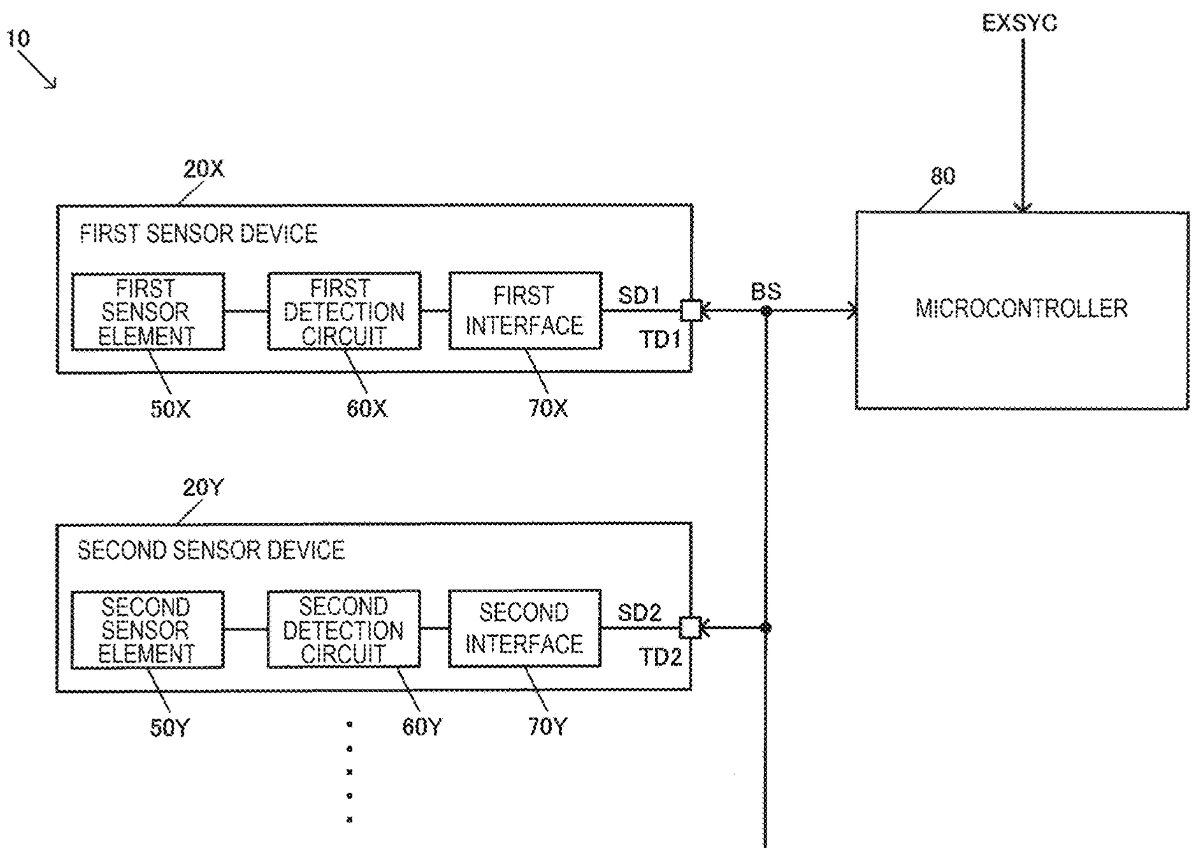


FIG. 1

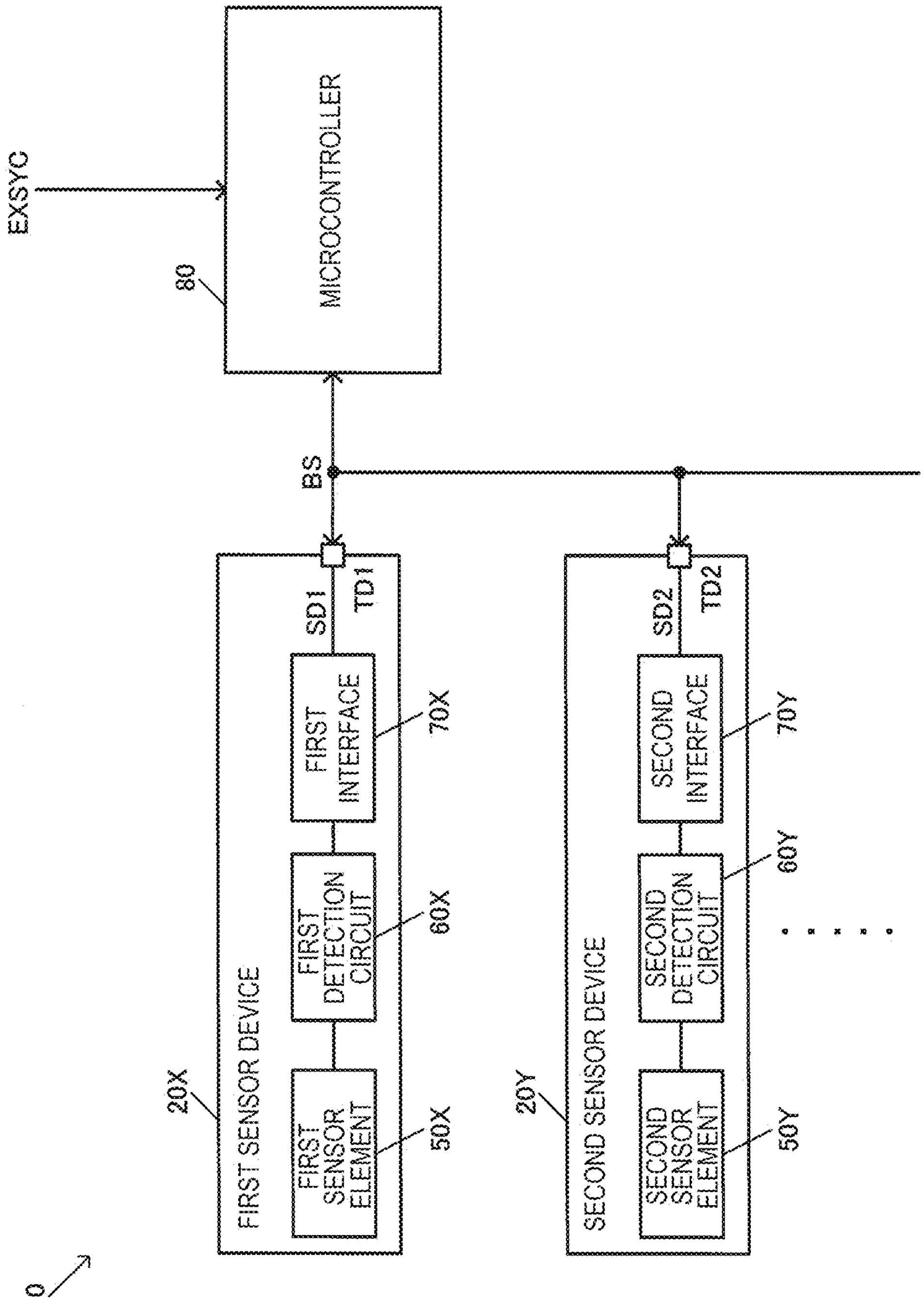

EXSYC
80
MICROCONTROLLER
BS
SD1
TD1
SD2
TD2
20X
FIRST SENSOR DEVICE
FIRST SENSOR ELEMENT
50X
FIRST DETECTION CIRCUIT
60X
FIRST INTERFACE
70X
20Y
SECOND SENSOR DEVICE
SECOND SENSOR ELEMENT
50Y
SECOND DETECTION CIRCUIT
60Y
SECOND INTERFACE
70Y
10

FIG. 2

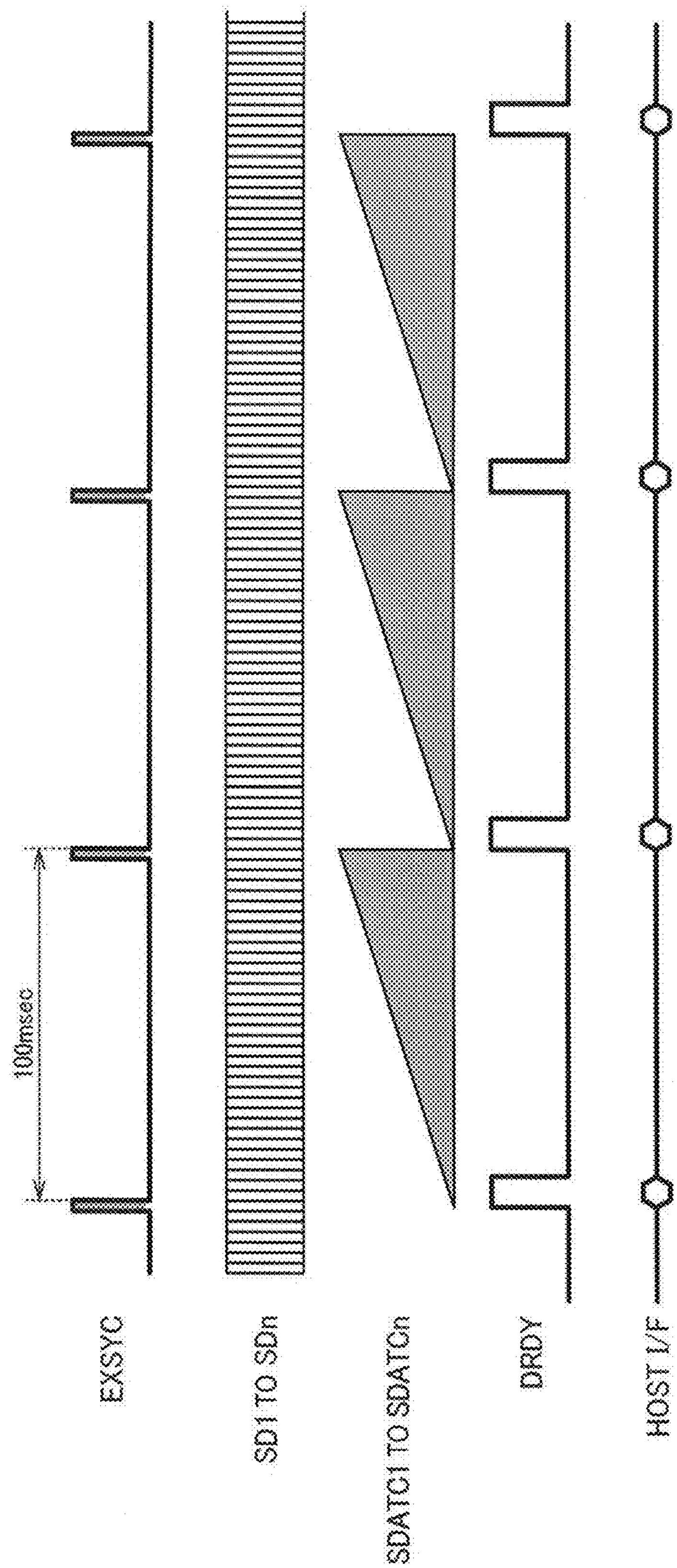
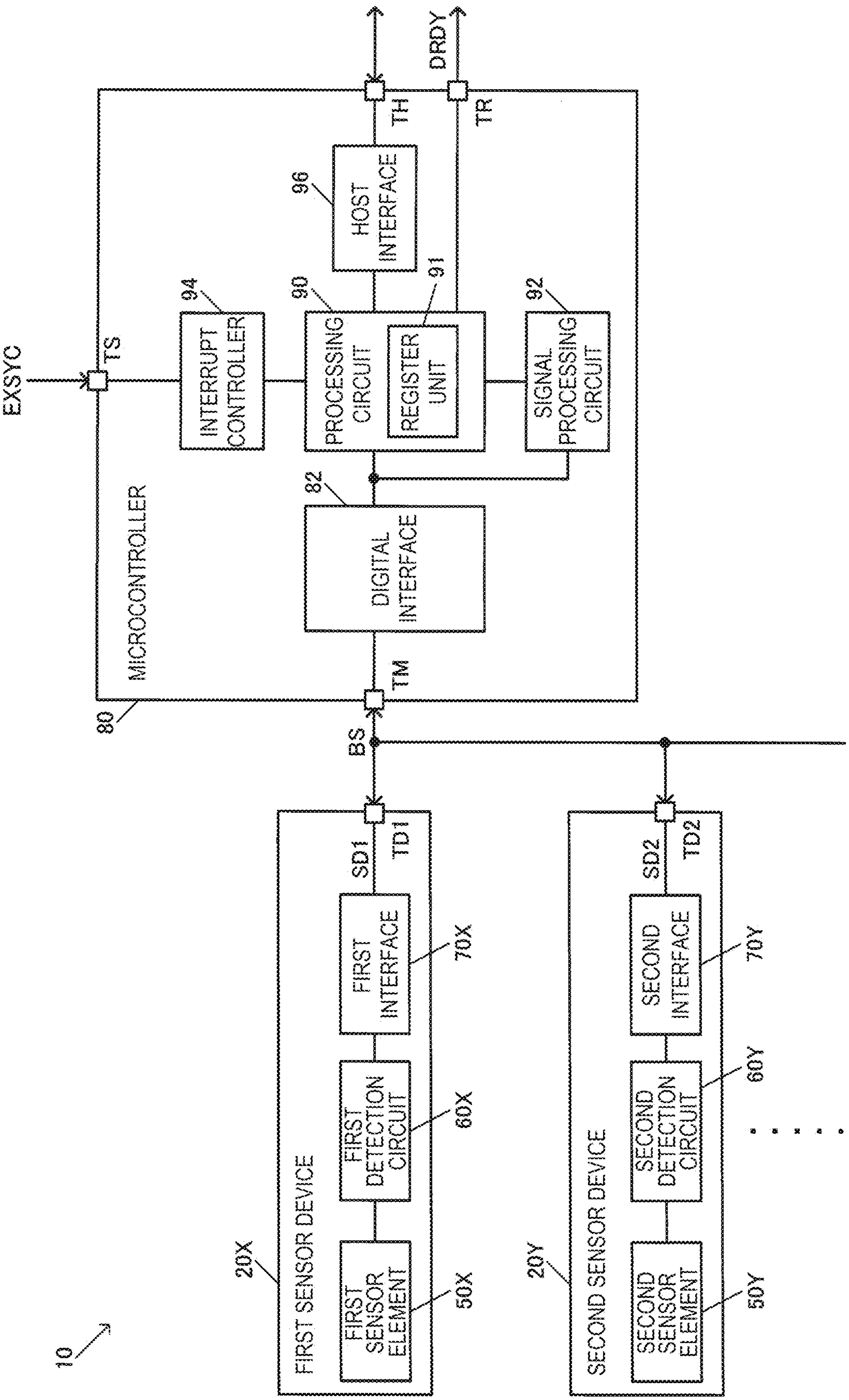


FIG. 3



EXSYC
TS
DRDY
TH
TR
MICROCONTROLLER
80
INTERRUPT CONTROLLER
94
PROCESSING CIRCUIT
90
REGISTER UNIT
91
HOST INTERFACE
96
SIGNAL PROCESSING CIRCUIT
92
DIGITAL INTERFACE
82
TM
BS
FIRST SENSOR DEVICE
20X
FIRST SENSOR ELEMENT
50X
FIRST DETECTION CIRCUIT
60X
FIRST INTERFACE
70X
SD1
TD1
SECOND SENSOR DEVICE
20Y
SECOND SENSOR ELEMENT
50Y
SECOND DETECTION CIRCUIT
60Y
SECOND INTERFACE
70Y
SD2
TD2
10

FIG. 4

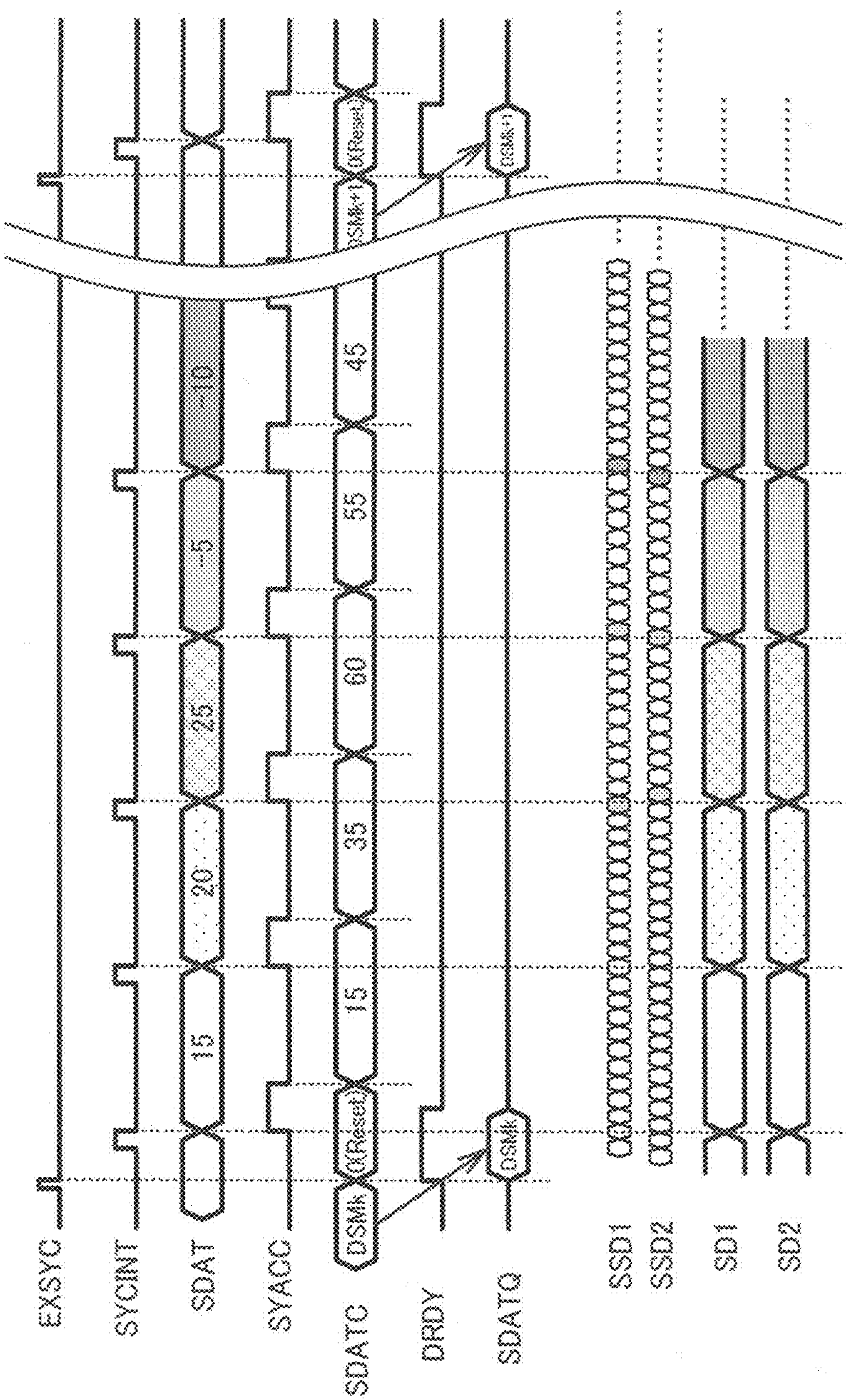

EXSYC
SYCINT
SDAT
SYACC
SDATC
DRDY
SDATQ
SSD1
SSD2
SD1
SD2
15
20
25
-5
-10
DSMk
0(Reset)
15
35
60
55
45
DSMk+1
0(Reset)
DSMk
DSMk+1

FIG. 5

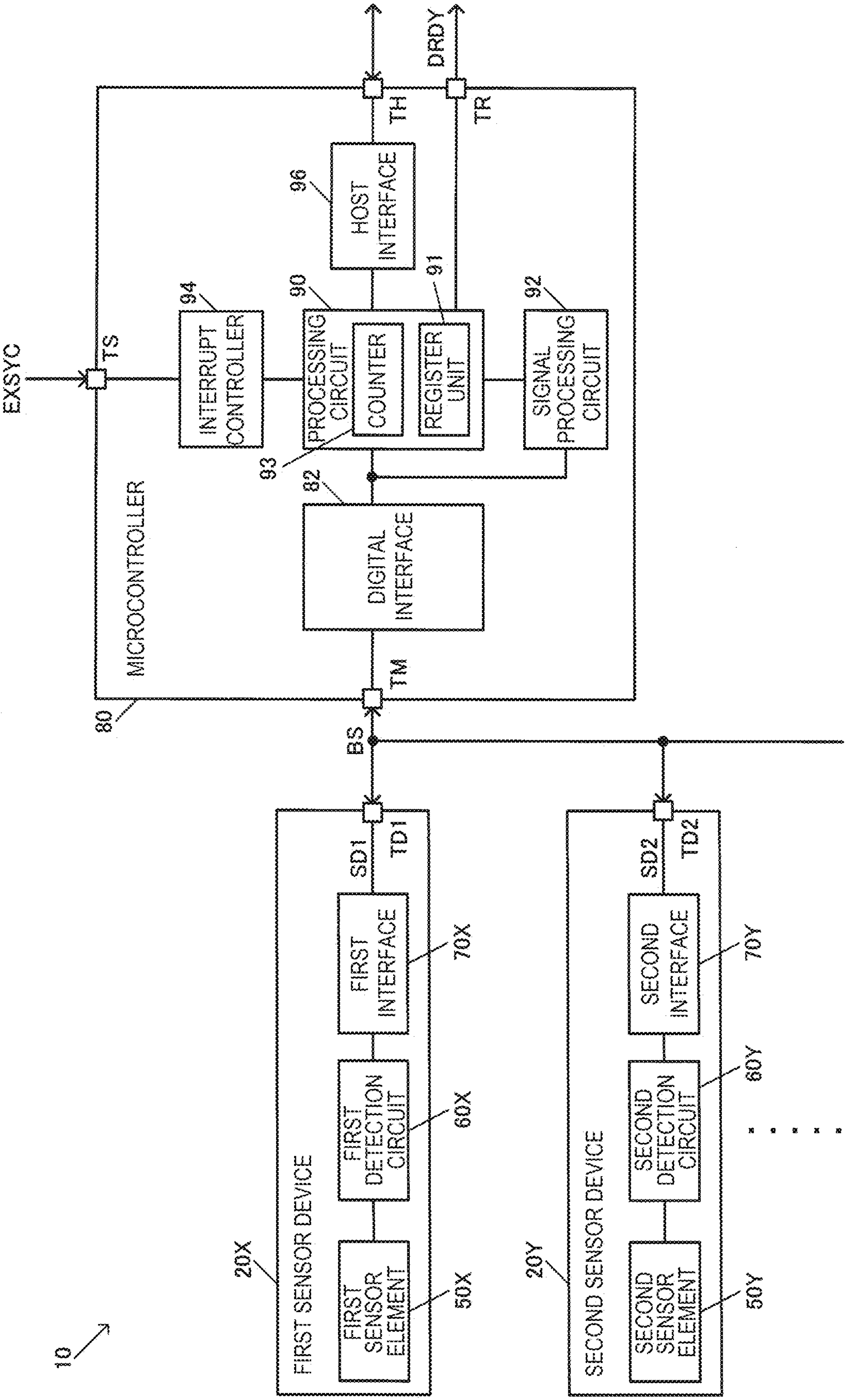

10
FIRST SENSOR DEVICE
20X
FIRST SENSOR ELEMENT
50X
FIRST DETECTION CIRCUIT
60X
FIRST INTERFACE
70X
SD1
TD1
SECOND SENSOR DEVICE
20Y
SECOND SENSOR ELEMENT
50Y
SECOND DETECTION CIRCUIT
60Y
SECOND INTERFACE
70Y
SD2
TD2
BS
80
MICROCONTROLLER
TM
DIGITAL INTERFACE
82
93
PROCESSING CIRCUIT
COUNTER
REGISTER UNIT
90
91
INTERRUPT CONTROLLER
94
SIGNAL PROCESSING CIRCUIT
92
HOST INTERFACE
96
EXSYC
TS
TH
TR
DRDY

FIG. 6

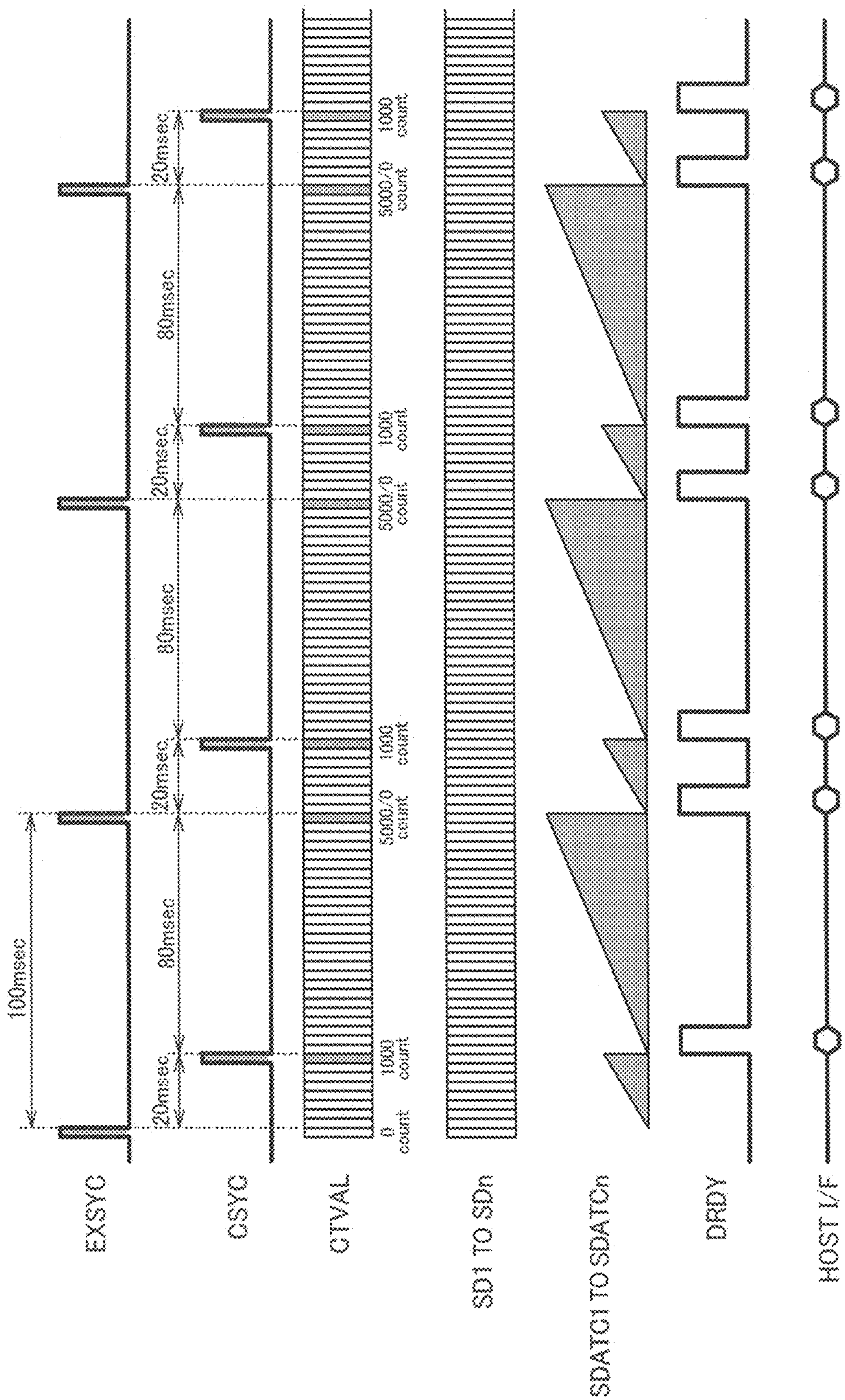

EXSYC
CSYC
CTVAL
SD1 TO SDn
SDATC1 TO SDATCn
DRDY
HOST I/F
100msec
20msec
80msec
20msec
80msec
20msec
80msec
20msec
0 count
1000 count
5000/0 count
1000 count
5000/0 count
1000 count
5000/0 count
1000 count

FIG. 7

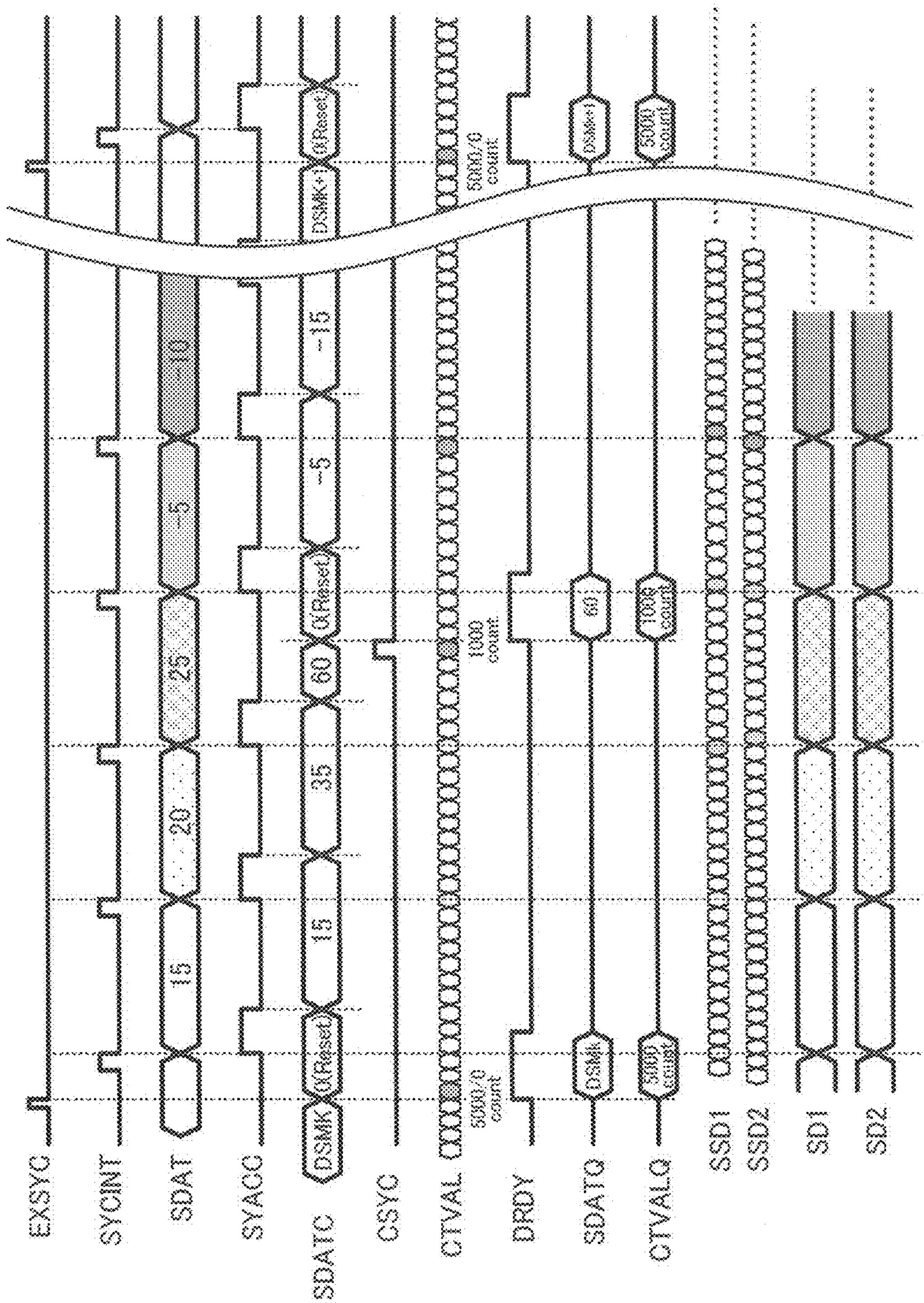

EXSYC
SYCINT
SDAT
SYACC
SDATC
CSYC
CTVAL
DRDY
SDATQ
CTVALQ
SSD1
SSD2
SD1
SD2
15
20
25
-5
-10
DSMK
0(Reset)
15
35
60
0(Reset)
-5
-15
DSMK+1
0(Reset)
5000/0 count
1000 count
5000/0 count
DSMk
60
DSMk+1
5000 count
1000 count
5000 count

FIG. 8

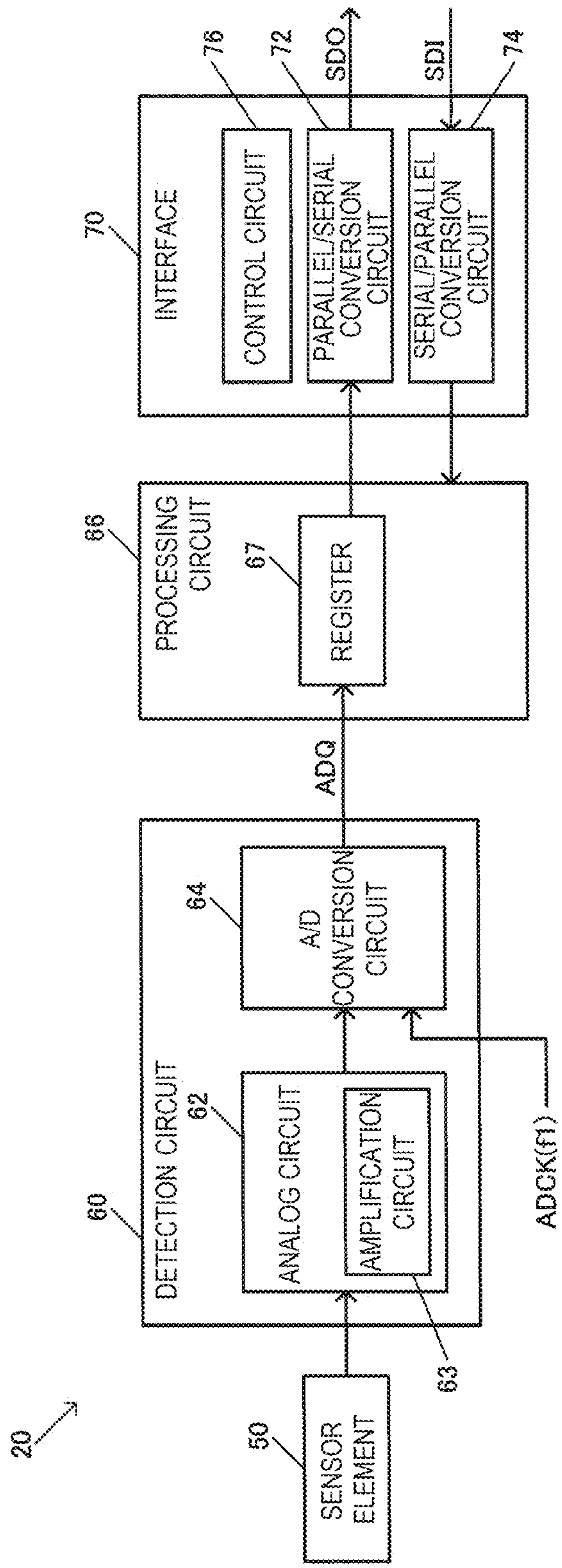

20
SENSOR ELEMENT
50
DETECTION CIRCUIT
60
ANALOG CIRCUIT
62
AMPLIFICATION CIRCUIT
63
A/D CONVERSION CIRCUIT
64
ADCK(f1)
ADQ
PROCESSING CIRCUIT
66
REGISTER
67
INTERFACE
70
CONTROL CIRCUIT
76
PARALLEL/SERIAL CONVERSION CIRCUIT
72
SERIAL/PARALLEL CONVERSION CIRCUIT
74
SDO
SDI

FIG. 9

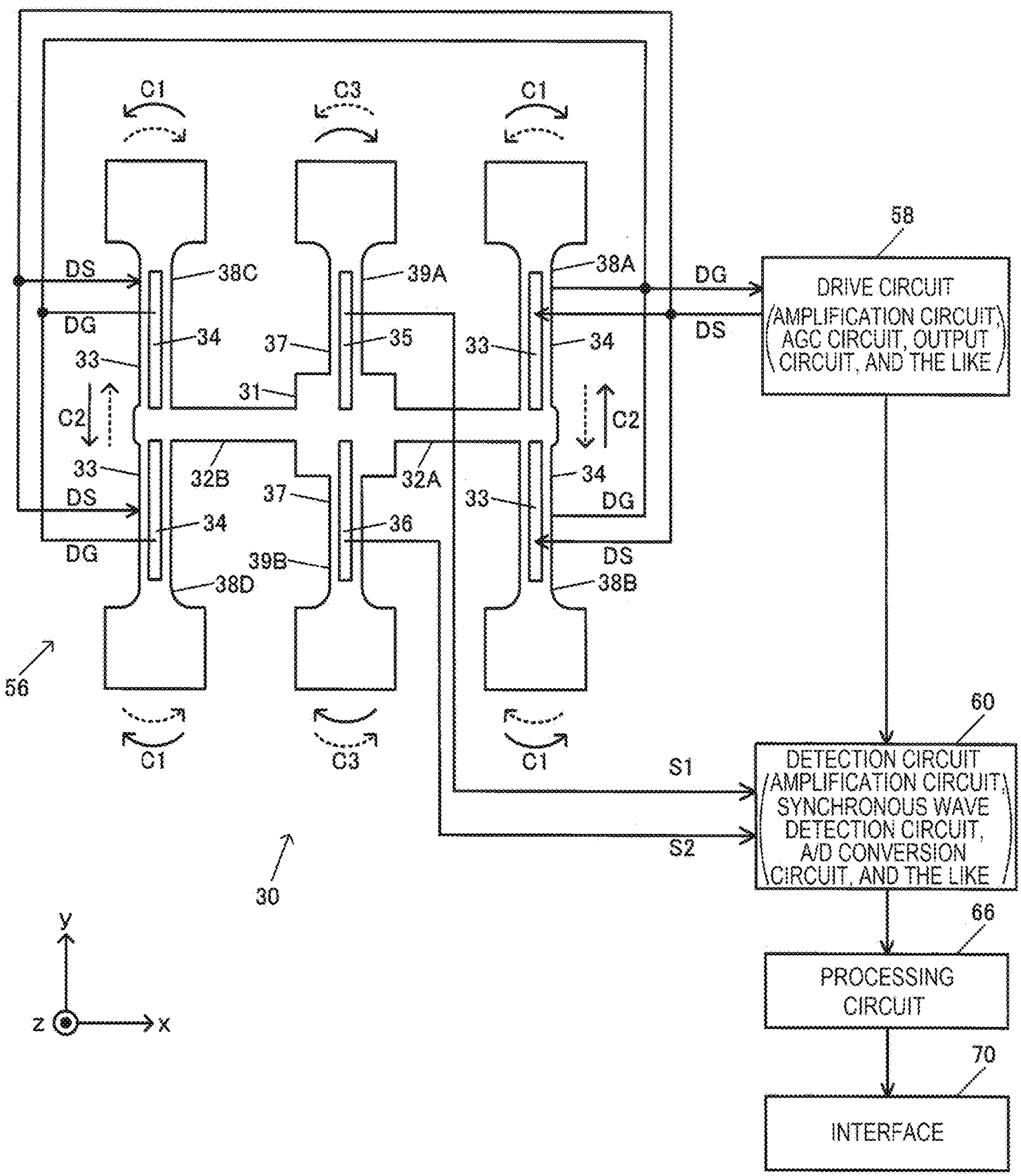

C1
C3
C1
58
38C
39A
38A
DS
DG
DRIVE CIRCUIT
(AMPLIFICATION CIRCUIT, AGC CIRCUIT, OUTPUT CIRCUIT, AND THE LIKE)
DG
DS
33
34
37
35
33
34
31
C2
C2
33
32B
32A
34
DS
37
33
DG
34
36
DG
39B
DS
38D
38B
56
C1
C3
C1
60
S1
DETECTION CIRCUIT
(AMPLIFICATION CIRCUIT, SYNCHRONOUS WAVE DETECTION CIRCUIT, A/D CONVERSION CIRCUIT, AND THE LIKE)
S2
30
66
y
PROCESSING CIRCUIT
z
x
70
INTERFACE

FIG. 10

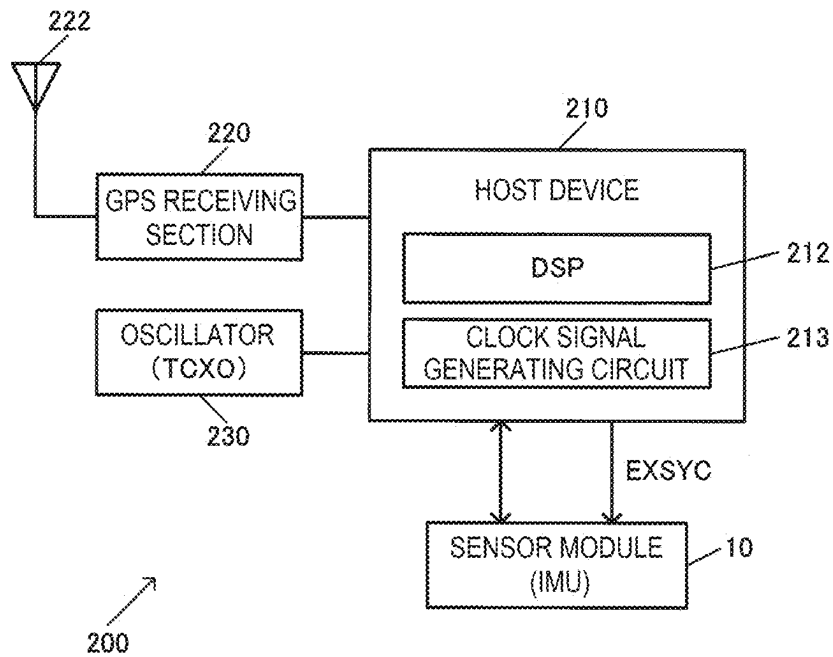
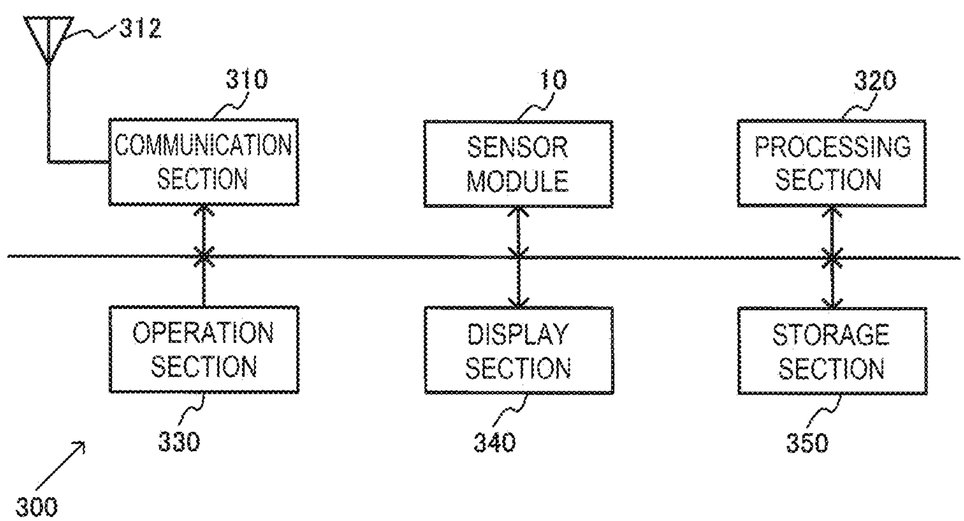

222
220
GPS RECEIVING SECTION
210
HOST DEVICE
DSP
212
CLOCK SIGNAL GENERATING CIRCUIT
213
OSCILLATOR (TCXO)
230
EXSYC
SENSOR MODULE (IMU)
10
200

FIG. 11


312
310
COMMUNICATION SECTION
10
SENSOR MODULE
320
PROCESSING SECTION
OPERATION SECTION
DISPLAY SECTION
STORAGE SECTION
330
340
350
300

SENSOR MODULE AND MEASUREMENT SYSTEM

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-036742, filed Mar. 11, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a sensor module, a measurement system, and the like.

2. Related Art

[0003] JP-A-2022-72338 discloses an inertia measuring apparatus. The inertia measuring apparatus includes an angular velocity sensor and an acceleration sensor that output inertia information, a storage that stores multiple correction parameters relating to the range of the value of the inertia information, a parameter control section that selects a selected correction parameter from the multiple correction parameters, and a correction operation section that corrects the inertia information by using the selected correction parameter.

[0004] JP-A-2022-072338 is an example of the related art.

[0005] The inertia measuring apparatus described above outputs angular velocity data and acceleration data, so that there is a problem of a large load on another apparatus that receives the data. For example, the communication between the inertia measuring apparatus and the other apparatus is performed at a high rate, so that there is a problem of a large communication load, or a large load on the other apparatus that processes the angular velocity data and the acceleration data.

SUMMARY

[0006] An aspect of the present disclosure relates to a sensor module including: a first sensor device including a first sensor element, a first detection circuit configured to receive a signal input from the first sensor element and perform detection, and a first interface configured to output first detected data from the first detection circuit; a second sensor device including a second sensor element, a second detection circuit configured to receive a signal input from the second sensor element and perform detection, and a second interface configured to output second detected data from the second detection circuit; and a microcontroller including a synchronization terminal to which an external synchronization signal is input and configured to receive the first detected data input from the first sensor device and the second detected data input from the second sensor device, the microcontroller configured to determine first integrated data by integrating the first detected data up to a synchronization timing of the external synchronization signal, determine second integrated data by integrating the second detected data up to the synchronization timing, and output the first integrated data and the second integrated data to a host device at the synchronization timing.

[0007] Another aspect of the present disclosure relates to a measurement system including the sensor module described above; and the host device electrically coupled to the sensor module.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** shows an exemplary configuration of a sensor module.

[0009] FIG. **2** is an exemplary timing chart illustrating the operation of a microcontroller.

[0010] FIG. **3** shows a first exemplary configuration of the microcontroller.

[0011] FIG. **4** is a signal waveform diagram for illustrating the operation of the sensor module.

[0012] FIG. **5** shows a second exemplary configuration of the microcontroller.

[0013] FIG. **6** is an exemplary timing chart for illustrating the operation of the microcontroller in the second exemplary configuration.

[0014] FIG. **7** is a signal waveform diagram for illustrating the operation of the sensor module using the microcontroller having the exemplary second configuration.

[0015] FIG. **8** shows an exemplary configuration of a sensor device.

[0016] FIG. **9** shows an exemplary configuration of an angular velocity sensor device.

[0017] FIG. **10** shows an exemplary configuration of a measurement system.

[0018] FIG. **11** shows an exemplary configuration of an electronic instrument.

DESCRIPTION OF EMBODIMENTS

[0019] A preferred embodiment of the present disclosure will be described below in detail. The present embodiment described below does not unduly limit the contents described in the claims, and all configurations described in the present embodiment are not necessarily essential configuration requirements of the present disclosure.

1. Exemplary Configuration of Sensor Module

[0020] FIG. **1** shows an exemplary configuration of a sensor module **10** according to the present embodiment. The sensor module **10** is a physical quantity detecting module configured with multiple sensor devices, and the sensor module **10** provides a sensor system or a sensor unit. The sensor module **10** shown in FIG. **1** includes a first sensor device **20**X, a second sensor device **20**Y, and a microcontroller **80**. Note that the sensor module **10** does not necessarily have the configuration shown in FIG. **1**, and may be variously modified by omitting some of the elements that constitute the sensor module **10** or adding other elements thereto. For example, FIG. **1** shows an exemplary case where the number of the sensor devices is two, but the number of the sensor devices may be three or more. For example, the sensor module **10** according to the present embodiment may include a first sensor device to an n-th sensor device. The symbol n is an integer greater than or equal to two.

[0021] The first sensor device **20**X includes a first sensor element **50**X, a first detection circuit **60***x*, which receives a signal input from the first sensor element **50**X and performs detection, and a first interface **70**X, which outputs first detected data SD**1** from the first detection circuit **60**X. The first sensor device **20**X is a device in which the first sensor element **50**X, and an integrated circuit unit including the first detection circuit **60***x* and the first interface **70**X are housed in a package. The integrated circuit unit is an IC chip realized by semiconductor.

[0022] The second sensor device **20**Y includes a second sensor element **50**Y, a second detection circuit **60**Y, which receives a signal input from the second sensor element **50**Y and performs detection, and a second interface **70**Y, which outputs second detected data SD**2** from the second detection circuit **60**Y. The second sensor device **20**Y is a device in which the second sensor element **50**Y, and an integrated circuit unit including the second detection circuit **60**Y and the second interface **70**Y are housed in a package.

[0023] When the number of the sensor devices is three or more, and the n-th sensor device is provided, the n-th sensor device includes an n-th sensor element, an n-th detection circuit that receives a signal input from the n-th sensor element and performs detection, and an n-th interface that outputs n-th detected data from the n-th detection circuit.

[0024] The first sensor element **50**X and the second sensor element **50**Y are each a sensor element that detects a physical quantity, and may be called a physical quantity transducer. The physical quantity is, for example, angular velocity, acceleration, angular acceleration, velocity, distance, pressure, sound pressure, and the amount of magnetism. The first sensor element **50**X and the second sensor element **50**Y detect physical quantities different from each other. For example, when the physical quantity is angular velocity, the first sensor element **50**X detects angular velocity around a first axis, and the second sensor element **50**Y detects angular velocity around a second axis. Instead, when the physical quantity is acceleration, the first sensor element **50**X detects acceleration in the direction of the first axis, and the second sensor element **50**Y detects acceleration in the direction of the second axis. Still instead, the first sensor element **50**X may detect a first physical quantity out of angular velocity, acceleration, angular acceleration, velocity, distance, pressure, and the amount of magnetism, which are physical quantities, and the second sensor element **50**Y may detect a second physical quantity different from the first physical quantity. As an example, the first sensor element **50**X detects angular velocity, and the second sensor element **50**Y detects acceleration.

[0025] The first detection circuit **60**X and the second detection circuit **60**Y can each include an analog circuit and an A/D conversion circuit that converts an analog signal from the analog circuit into digital data. The analog circuit may include an amplification circuit that amplifies a signal from the sensor element, a wave detecting circuit such as a synchronous wave detecting circuit, a gain adjusting circuit, an offset adjusting circuit, or the like. The A/D conversion circuit outputs digital detected data to the interface. A/D conversion circuits classified to an A/D conversion type may include, for example, those of a successive comparison type, a delta-sigma type, a flash type, a pipeline type, and a double integration type.

[0026] The first interface **70**X and the second interface **70**Y are each, for example, a circuit that performs digital interfacing, for example, transmitting or receiving serial data. Specifically, the interfaces perform interfacing that complies with the SPI or I2C communication standard. The interfaces may perform interfacing that complies with an advanced SPI or I2C communication standard, or a partially improved or modified SPI or I2C communication standard.

[0027] The first detected data SD**1** from the first sensor device **20**X and the second detected data SD**2** from the second sensor device **20**Y are input to the microcontroller **80**. The sensor module **10** includes a digital interface bus BS, which electrically couples the first sensor device **20**X and the second sensor device **20**Y to the microcontroller **80**. The digital interface bus BS is a bus that complies with a communication standard in accordance with which the first interface **70**X and the second interface **70**Y perform the interfacing. The digital interface bus BS includes a data signal line, a clock signal line, and other lines. The digital interface bus BS may further carry a chip select signal. The microcontroller **80** receives the first detected data SD**1** input from the first sensor device **20**X and the second detected data SD**2** input from the second sensor device **20**Y via the digital interface bus BS. Note that the first interface **70**X and that the second interface **70**Y are electrically coupled to the digital interface bus BS via a terminal TD**1** and a terminal TD**2**, respectively. The term "electrically coupled" used herein means coupling that allows an electrical signal transmission and information transmission through the electrical signal. The microcontroller **80** is a controller that serves as a master with respect to the first sensor device **20**X and the second sensor device **20**Y. The microcontroller **80** is an integrated circuit unit, and can be realized, for example, by a processor such as an MPU or a CPU. The microcontroller **80** may instead be an ASIC or an FPGA.

[0028] An external synchronization signal EXSYC is input to the microcontroller **80**. The external synchronization signal EXSYC is a signal input from an external device to the sensor module **10** and becomes active at each synchronization timing. The external synchronization signal EXSYC is, for example, a signal that becomes active at regular intervals. The term "active" is a high level in the case of positive logic and a low level in the case of negative logic. For example, the external device may be a receiver of a satellite positioning system, and a PPS signal from the receiver may be input as the external synchronization signal EXSYC to the sensor module **10**. Instead, the external device may be a host device for the sensor module **10**, and the host device may generate the external synchronization signal EXSYC synchronized with the PPS signal, for example, by using a PLL. Still instead, the host device may acquire time information, for example, from an NTP service via a network, and generate the external synchronization signal EXSYC synchronized with the time information.

[0029] FIG. **2** is an exemplary timing chart illustrating the operation of the microcontroller **80**. FIG. **2** shows a case where a 10-pulse/sec PPS signal is input as the external synchronization signal EXSYC to the microcontroller **80**.

[0030] In the waveform of the first detected data SD**1** to n-th detected data SDn, one quadrangle indicates one set of sampled data. The term "sampling" used herein means that the microcontroller **80** receives detected data from the interface of each of the sensor devices. Note that FIG. **2** shows only one waveform, but the microcontroller **80** samples each of the first detected data SD**1** to the n-th detected data SDn.

[0031] The microcontroller **80** integrates the first detected data SD**1** to determine first integrated data SDATC**1**. Similarly, the microcontroller **80** integrates the second detected data SD**2** to the n-th detected data SDn to determine second integrated data SDATC**2** to n-th integrated data SDATCn. FIG. **2** shows only the waveform of one set of integrated data in a case where the detected data is a constant value. The microcontroller **80** resets the first integrated data SDATC**1** to the n-th integrated data SDATCn at an edge timing of the external synchronization signal EXSYC, and continues the integration until the edge timing of the next external synchronization signal EXSYC. The edge timing of the external synchronization signal EXSYC may be either the rising or falling timing of the external synchronization signal EXSYC.

[0032] The microcontroller **80** outputs the first integrated data SDATC**1** to the n-th integrated data SDATCn as the result of the integration performed from the edge timing of the external synchronization signal EXSYC to the next edge timing thereof to the host device. Specifically, the microcontroller **80** holds the first integrated data SDATC**1** to the n-th integrated data SDATCn immediately before reset at the edge timing of the external synchronization signal EXSYC, and outputs the held first integrated data SDATC**1** to n-th integrated data SDATCn to the host device. A host interface of the microcontroller **80** is an interface that complies, for example, with SPI, I2C, or UART. When the host interface is an interface that complies with SPI, the microcontroller **80** generates a data ready signal DRDY at the edge timing of the external synchronization signal EXSYC and outputs the data ready signal DRDY to the host device. Upon reception of the data ready signal DRDY, the host device reads the first integrated data SDATC**1** to the n-th integrated data SDATCn via the host interface of the microcontroller **80**. Note that when the host interface is an interface that complies with I2C or UART, the data ready signal DRDY may be generated in the microcontroller **80** although the data ready signal DRDY is not used for communication.

[0033] FIG. **3** shows a first exemplary configuration of the microcontroller **80**. The microcontroller **80** includes a digital interface **82**, a processing circuit **90**, a signal processing circuit **92**, an interrupt controller **94**, and a host interface **96**. The number of the sensor devices is set at n=2 in the description.

[0034] The digital interface **82** is a circuit that performs interfacing with the sensor devices. That is, the digital interface **82** performs interfacing as a master with respect to the first interface **70**X and the second interface **70**Y. The digital interface **82** is coupled to the digital interface bus BS via a terminal TM. The digital interface **82** performs interfacing that complies with the SPI or I2C communication standard, an advanced SPI or I2C communication standard, or a partially improved or modified SPI or I2C communication standard, as the first interface **70**X and the second interface **70**Y.

[0035] The signal processing circuit **92** is a circuit that performs digital signal processing such as the integration described above, and can be realized, for example, by a DSP. The signal processing circuit **92** integrates the first detected data SD**1** received by the digital interface **82** to determine the first integrated data SDATC**1**, and integrates the second detected data SD**2** received by the digital interface **82** to determine the second integrated data SDATC**2**. The signal processing circuit **92** may perform filtering or correction. Specifically, the signal processing circuit **92** may calculate the moving average of the latest J pieces of detected data, and then perform 1/K-rate down-sampling on the resultant detected data (J and K are each integer greater than or equal to two). The signal processing circuit **92** may further perform correction, such as temperature correction, on the filtered detected data. The signal processing circuit **92** may integrate the filtered or the corrected detected data to determine integrated data.

[0036] The processing circuit **90** is a circuit corresponding to a core CPU of the microcontroller **80**, and performs various types of operation or control. The processing circuit **90** includes a register unit **91** including various registers. The processing circuit **90** saves the integrated data from the signal processing circuit **92** in the register unit **91**. The processing circuit **90** then generates the signal DRDY, which is an integrated data preparation completion signal, and outputs the signal DRDY to a host device **210**, which will be described later in FIG. **10**, via a terminal TR. The signal DRDY is a signal indicating that the signal processing circuit **92** has completed the digital signal processing. Note that when the host interface **96** performs communication that complies with the UART standard, the signal DRDY may not be output to the host device **210**.

[0037] The register unit **91** includes multiple registers that can be accessed externally. For example, the host device **210** can access a data register of the register unit **91** via the host interface **96**, and read the integrated data. The processing circuit **90** may count the frequency indicating how frequently the data register for the integrated data has been updated, and write the counted update frequency to an update frequency register of the register unit **91**. The host device **210** can thus identify how many sets of integrated data have been read from the microcontroller **80**.

[0038] The interrupt controller **94** accepts various interrupt requests. The interrupt controller **94** then outputs signals that notify an interrupt request, an interrupt level, and a vector number to the processing circuit **90** in accordance with the priority and the interrupt level. The external synchronization signal EXSYC is input as one of the interrupt request signals to the interrupt controller **94** via a synchronization terminal TS. Upon acceptance of the interrupt request using the external synchronization signal EXSYC, the processing circuit **90** performs interrupt corresponding to the interrupt request. Note that examples of the interrupt request include an SPI or UART interrupt request from the host interface **96**, interrupt requests from various timers, an I2C interrupt request.

[0039] The host interface **96** is a circuit that performs digital interfacing with the host device **210**. For example, the host interface **96** performs serial data communication that complies, for example, with SPI, I2C, or UART as host interfacing.

[0040] FIG. **4** is a signal waveform diagram for illustrating the operation of the sensor module **10**. The number of the sensor devices is set at n=2 in the description. Note that data hatched in the same manner indicates data corresponding to each other.

[0041] The symbol SSD**1** shown in FIG. **4** indicates detected data output by the first detection circuit **60**X of the first sensor device **20**X, and the symbol SSD**2** indicates detected data output from the second detection circuit **60**Y of the second sensor device **20**Y. In the present embodiment, the first sensor device **20**X and the second sensor device **20**Y operate based on separate clock signals. For example, the sensor devices each operate based on a clock signal from an oscillation circuit built in the sensor device or a clock signal generated by using a vibrator such as a quartz crystal vibrator provided in the sensor device. The detected data SSD**1** output by the first detection circuit **60**X and the detected data SSD**2** output from the second detection circuit **60**Y are therefore out of synch with each other.

[0042] The interrupt controller **94** generates at regular intervals a signal SYCINT used to perform internal interrupt. The frequency of the signal SYCINT is lower than the rates at which the detected data SSD**1** and SSD**2** are sampled. The microcontroller **80** starts the internal interrupt at an edge timing of the signal SYCINT. The following description will be made with reference to a case where the internal interrupt starts at the falling timing of the signal SYCINT.

[0043] When the internal interrupt starts, the digital interface **82** receives the first detected data SD**1** from the first interface **70**X of the first sensor device **20**X. The first detected data SD**1** is data at the falling timing of the signal SYCINT out of the detected data SSD**1** output by the first detection circuit **60**X. The digital interface **82** further receives the second detected data SD**2** from the second interface **70**Y of the second sensor device **20**Y. The second detected data SD**2** is data at the falling timing of the signal SYCINT out of the detected data SSD**2** output by the second detection circuit **60**Y. Detected data SDAT received by the digital interface **82** is the first detected data SD**1** or the second detected data SD**2**. Only one set of detected data is shown in FIG. **4**, but the digital interface **82** receives each of the first detected data SD**1** and the second detected data SD**2**.

[0044] When the internal interrupt starts, the processing circuit **90** generates a signal SYACC, which causes the signal processing circuit **92** to perform operation. FIG. **4** shows a case where the signal SYACC becomes active for a predetermined period from the falling edge of the signal SYCINT. It is assumed in the description that the signal SYACC is active when it has a high level. In each period for which the signal SYACC is active, the signal processing circuit **92** adds the detected data SDAT to integrated data SDATC at that point of time to update the integrated data SDATC in accordance with the result of the addition. The update is performed, for example, when the period for which the signal SYACC is active ends. The detected data SDAT is added whenever the internal interrupt is performed to generate the integrated data SDATC. Although FIG. **4** shows only one set of integrated data, the first detected data SD**1** and the second detected data SD**2** are each integrated, so that the first integrated data and the second integrated data are generated.

[0045] The interrupt controller **94** generates an external interrupt signal at an edge timing of the external synchronization signal EXSYC, and the microcontroller **80** starts external interrupt in response to the external interrupt signal. FIG. **4** shows a case where the external interrupt starts at the falling timing of the external synchronization signal EXSYC. When the external interrupt starts, the signal processing circuit **92** resets the integrated data SDATC to zero. The detected data SDAT is thus integrated from the edge timing of the external synchronization signal EXSYC to the next edge timing thereof. When the value of the detected data SDAT after the external synchronization signal EXSYC is input is 15, 20, 25, −5, −10, . . . , the value of the integrated data SDATC is 0, 0+15=15, 15+20=35, 35+25=60, 60−5=55, 55−10=45, . . . , as shown in FIG. **4**. The integrated data SDATC immediately before reset at the edge timing of the external synchronization signal EXSYC is called DSMk and DSMk+1. The symbol k is an integer and is an index indicating how many edges of the external synchronization signal EXSYC have been used to generate the integrated data.

[0046] When the integrated data SDATC is reset, the processing circuit **90** saves the integrated data DSMk and DSMk+1 immediately before the reset operation in the register unit **91**, and outputs the data ready signal DRDY to the host device **210**. The host device **210** receives the data ready signal DRDY and issues a read request to the host interface **96**, and the host interface **96** outputs the integrated data DSMk and DSMk+1 stored in the register unit **91** to the host device **210** as output data SDATQ. Note that when the host interface **96** performs communication that complies with the UART standard, the integrated data DSMk and DSMk+1 may be output to the host device **210** with the signal DRDY not output to the host device **210**.

[0047] In the present embodiment, the sensor module **10** includes the first sensor device **20**X, the second sensor device **20**Y, and the microcontroller **80**. The first sensor device **20**X includes the first sensor element **50**X, the first detection circuit **60**X, which receives a signal input from the first sensor element **50**X and performs detection, and the first interface **70**X, which outputs the first detected data SD**1** from the first detection circuit **60**X. The second sensor device **20**Y includes the second sensor element **50**Y, the second detection circuit **60**Y, which receives a signal input from the second sensor element **50**Y and performs detection, and the second interface **70**Y, which outputs the second detected data SD**2** from the second detection circuit **60**Y. The microcontroller **80** includes the synchronization terminal TS, to which the external synchronization signal EXSYC is input. The microcontroller **80** receives the first detected data SD**1** input from the first sensor device **20**X and the second detected data SD**2** input from the second sensor device **20**Y. The microcontroller **80** determines the first integrated data SDATC**1** by integrating the first detected data SD**1** up to the synchronization timing of the external synchronization signal EXSYC, determines the second integrated data SDATC**2** by integrating the second detected data SD**2** up to the synchronization timing, and outputs the first integrated data SDATC**1** and the second integrated data SDATC**2** to the host device at the synchronization timing.

[0048] According to the present embodiment, the data rate at which the integrated data is integrated is lower than the rate at which the detected data is sampled. The configuration in which the microcontroller **80** outputs the integrated data to the host device therefore reduces the load on the host device as compared with a configuration in which the detected data is output. For example, the communication load or the operation load such as the integration on the host device decreases. The host device performs, for example, communication with or control on various sensors or apparatuses provided in the system, reducing the load on the sensor module **10** allows the processing load to be allocated to other sensors or apparatuses.

[0049] Furthermore, according to the present embodiment, the microcontroller **80** outputs the integrated data to the host device in synchronization with the external synchronization signal EXSYC. The host device can thus acquire synchronized data, for example, from various sensors. That is, the host device can receive data from the sensor module **10**, other sensors, and the like in synchronization with the external synchronization signal EXSYC, and can perform various types of processing, control, or other operation by using the synchronized data. For example, in autonomous navigation of a moving object, the position and posture of the moving object can be detected by using various synchronized sensor outputs.

[0050] In the present embodiment, the microcontroller **80** may determine the first: integrated data SDATC**1** by integrating the first detected data SD**1** for the period between the synchronization timings of the external synchronization signal EXSYC, and may determine the second integrated data SDATC**2** by integrating the second detected data SD**2** for the period between the synchronization timings.

[0051] In the present embodiment, the microcontroller **80** may reset the first integrated data SDATC**1** and the second integrated data SDATC**2** whenever the synchronization timing is reached.

[0052] According to the present embodiment, the host device can acquire data integrated for each period between the synchronization timings of the external synchronization signal EXSYC. The host device can use the integrated data regarded as the detected data from the sensor module **10**, or can further perform integration or other processing on the integrated data and use the resultant data. For example, when the sensor device is an angular velocity sensor and the external synchronization signal EXSYC is PPS signal, the integrated data indicates the angle of rotation per second. That is, the integrated data can be regarded as angular velocity data expressed in dps. The host device can, for example, use the received integrated data as angular velocity data, or can further integrate the angular velocity data to determine the angle of rotation.

[0053] Furthermore, in the present embodiment, the microcontroller **80** may output a signal that notifies completion of the integration for the period between the synchronization timings to an external apparatus when the integration is completed. In the examples shown in FIGS. **2** and **4**, the data ready signal DRDY is the "signal that notifies completion of the integration".

[0054] According to the present embodiment, upon reception of the signal that notifies completion of the integration, the host device can perform communication for receiving the integrated data, and acquire the result of the completed integration.

[0055] In the present embodiment, the external synchronization signal EXSYC may be a time reference signal acquired from a satellite positioning system. The satellite positioning system is also called a global navigation satellite system (GNSS), and is, for example, a global positioning system (GPS), a quasi-zenith satellite system (QZSS), GLONASS, and Galileo.

[0056] According to the present embodiment, the microcontroller **80** can output the integrated data to the host device in synchronization with the time reference signal. For example, in a system that performs processing, control, or other operation in synchronization with the time reference signal, the host device can acquire data synchronized with the time reference signal from the sensor module **10** and perform the processing, control, or other operation.

2. Second Exemplary Configuration

[0057] FIG. **5** shows a second exemplary configuration of the microcontroller **80**. The elements having already been described have the same reference characters, and will not be described as appropriate. Portions of the second exemplary configuration that differ from those of the first exemplary configuration will be primarily described below. In the second exemplary configuration, the processing circuit **90** includes a counter **93**.

[0058] The counter **93** starts counting a predetermined period from the synchronization timing of the external synchronization signal EXSYC. The microcontroller **80** includes an oscillation circuit that is not shown, and the counter **93** uses a clock signal from the oscillation circuit to perform the counting. Instead, the microcontroller **80** may receive a clock signal input from an external oscillation circuit, and the counter **93** may use the clock signal to perform the counting.

[0059] The register unit **91** stores set information indicating the predetermined period. The set information is, for example, information indicating a count value corresponding to the predetermined period. For example, the host device writes the set information to the register unit **91** via the host interface **96**. The processing circuit **90** compares the count value of the counter **93** with the set information stored in the register unit **91** to determine whether the predetermined period has been counted.

[0060] FIG. **6** is an exemplary timing chart for illustrating the operation of the microcontroller **80** in the second exemplary configuration. FIG. **6** shows a case where the 10-pulses/sec PPS signal is input as the external synchronization signal EXSYC to the microcontroller **80**, the counter **93** counts at 50 kHz, and the predetermined period corresponds to 1,000 counts.

[0061] The processing circuit **90** resets a count value CTVAL of the counter **93** to zero at the edge timing of the external synchronization signal EXSYC. The counter **93** increments the count value CTVAL based on the 50-kHz clock signal. The processing circuit **90** generates an internal synchronization signal CSYC when the count value CTVAL coincides with 1,000 counts. The signal processing circuit **92** resets the integrated data SDATC**1** to SDATCn at the edge timing of the external synchronization signal EXSYC and the edge timing of the internal synchronization signal CSYC. The signal processing circuit **92** integrates the detected data SD**1** to SDn for the period from the edge timing of the external synchronization signal EXSYC to the edge timing of the internal synchronization signal CSYC and for the period from the edge timing of the internal synchronization signal CSYC to the edge timing of the external synchronization signal EXSYC to generate the integrated data SDATC**1** to SDATCn.

[0062] The microcontroller **80** generates the data ready signal DRDY at each of the edge timing of the external synchronization signal EXSYC and the edge timing of the internal synchronization signal CSYC, and outputs the data ready signal DRDY to the host device. Upon reception of the data ready signal DRDY, the host device reads the first integrated data SDATC**1** to the n-th integrated data SDATCn via the host interface of the microcontroller **80**.

[0063] FIG. **7** is a signal waveform diagram for illustrating the operation of the sensor module **10** using the microcontroller **80** having the exemplary second configuration. The number of the sensor devices is set at n=2 in the description. Processes to be carried out before the microcontroller **80** receives the detected data SDAT from each of the sensor devices are the same as those in FIG. **4**.

[0064] The interrupt controller **94** generates the external interrupt signal at the edge timing of the external synchronization signal EXSYC, and the microcontroller **80** starts the external interrupt in response to the external interrupt signal. FIG. **7** shows a case where the external interrupt process starts at the falling timing of the external synchronization signal EXSYC. When the external interrupt starts, the signal processing circuit **92** resets the integrated data SDATC to zero. Thereafter, in the period for which the signal SYACC is active, the signal processing circuit **92** adds the detected data SDAT to the integrated data SDATC.

[0065] The processing circuit **90** generates the internal synchronization signal CSYC when the count value CTVAL coincides with 1,000 counts, which indicates the predetermined period. The signal processing circuit **92** resets the integrated data SDATC to zero at an edge timing of the internal synchronization signal CSYC. FIG. **7** shows a case where the integrated data SDATC is reset at the falling timing of the internal synchronization signal CSYC. Thereafter, in the period for which the signal SYACC is active, the signal processing circuit **92** adds the detected data SDAT to the integrated data SDATC.

[0066] As described above, the detected data SDAT is integrated for the period from the edge timing of the external synchronization signal EXSYC to the edge timing of the internal synchronization signal CSYC and for the period from the edge timing of the internal synchronization signal CSYC to the edge timing of the external synchronization signal EXSYC. It is assumed that the value of the detected data SDAT after the external synchronization signal EXSYC is input is 15, 20, 25, −5, −10, . . . , and the internal synchronization signal CSYC is generated when SDAT=25, as shown in FIG. **7**. In this case, the value of the integrated data SDATC is 0, 0+15=15, 15+20=35, and 35+25=60, and is reset by the internal synchronization signal CSYC, to 0, 0−5=−5, −5−10=−15, The integrated data SDATC immediately before reset at the edge timing of the external synchronization signal EXSYC and the edge timing of the internal synchronization signal CSYC is called DSMk, **60**, and DSMk+1. The symbol k is an integer and is an index indicating how many edges of the external synchronization signal EXSYC have been used to generate the integrated data.

[0067] When the integrated data SDATC is reset, the processing circuit **90** saves the integrated data DSMk, **60**, and DSMk+1 immediately before the reset operation and count values 5,000, 1,000, and 5,000 in the register unit **91**, and outputs the data ready signal DRDY to the host device **210**. The host device **210** receives the data ready signal DRDY and issues the read request to the host interface **96**, and the host interface **96** outputs, to the host device **210**, the integrated data DSMk, **60**, and DSMk+1 stored in the register unit **91** as the output data SDATQ and the count values 5,000, 1,000, and 5,000 as output data CTVALQ. Note that when the host interface **96** performs communication that complies with the UART standard, the integrated data DSMk, **60**, and DSMk+1 and the count values 5,000, 1,000, and 5,000 may be output to the host device **210** with the signal DRDY not output to the host device **210**.

[0068] In the present embodiment, the microcontroller **80** determines the first integrated data SDATC**1** by integrating the first detected data SD**1** for the period between output timings including the synchronization timing of the external synchronization signal EXSYC and the timing after the predetermined period from the synchronization timing, and may determine the second integrated data SDATC**2** by integrating the second detected data SD**2** for the period between the synchronization timings.

[0069] According to the present embodiment, the microcontroller **80** can output the integrated data to the host device at any output timing as well as the synchronization timing of the external synchronization signal EXSYC. The host device may acquire data from a sensor or the like at various synchronization timings, in which case, the host device can acquire the integrated data from the sensor module **10** at the various synchronization timings, and can perform various types of processing, control, or other operation by using the synchronized data.

[0070] In the present embodiment, the sensor module **10** may include a register that stores the set information on the predetermined period. The microcontroller **80** may set a timing after the predetermined period based on the set information.

[0071] According to the present embodiment, the timing when the integrated data is output can be set at any timing by writing the set information to the register from an apparatus external to the sensor module **10**.

[0072] In the present embodiment, the microcontroller **80** may reset the first integrated data SDATC**1** and the second integrated data SDATC**2** at each output timing.

[0073] According to the present embodiment, the integrated data is reset at each output timing, thus, the integrated data is data in which the detected data is integrated between the output timings. The host device can thus acquire data integrated for each period between output timings. The host device can use the integrated data regarded as the detected data from the sensor module **10**, or can further perform integration or other processing on the integrated data and use the resultant data. For example, when the sensor device is an angular velocity sensor, the integrated data can be converted into an angle of rotation per second based on the integrated data and the period between the output timings, and used as angular velocity data expressed in dps. The host device can instead also determine the angle of rotation by further integrating the integrated data.

[0074] Furthermore, in the present embodiment, the microcontroller **80** may output a signal that notifies completion of the integration for the period between the output timings to an external apparatus when the integration is completed. In the examples shown in FIGS. **6** and **7**, the data ready signal DRDY is the "signal that notifies completion of the integration".

[0075] According to the present embodiment, upon reception of the signal that notifies completion of the integration, the host device can perform communication for receiving the integrated data, and acquire the result of the completed integration.

3. Sensor Device

[0076] FIG. **8** shows an exemplary configuration of a sensor device **20**. The sensor device **20** corresponds to each of the first sensor device **20**X and the second sensor device **20**Y shown in FIGS. **1**, **3**, and **5**. The sensor device **20** includes a sensor element **50**, a detection circuit **60**, a processing circuit **66**, and an interface **70**.

[0077] The detection circuit **60** includes an analog circuit **62** including an amplification circuit **63**, which amplifies a signal from the sensor element **50**, and an A/D conversion circuit **64**, which converts an analog signal from the analog circuit **62** into digital data. The processing circuit **66**

includes a register **67**. The interface **70** includes a parallel/serial conversion circuit **72**, a serial/parallel conversion circuit **74**, and a control circuit **76**, which performs interface control.

[0078] The A/D conversion circuit **64** samples the analog detection signal from the analog circuit **62** based on a clock signal ADCK having a frequency f**1** to perform A/D conversion. The A/D conversion circuit **64** outputs detected data ADQ at an output sampling rate corresponding to the frequency f**1**. The register **67** holds the detected data ADQ. The detected data ADQ output at the frequency f**1** corresponds to the detected data SSD**1** and SSD**2** in FIGS. **4** and **7**. When the A/D conversion resolution of the A/D conversion circuit **64** is k bits, the detected data ADQ is, for example, k-bit parallel data.

[0079] Serial data carried by a data input signal SD**1** from the microcontroller **80** is converted into parallel data by the serial/parallel conversion circuit **74**. The microcontroller **80** transmits a read command to the interface **70** at the edge of the signal SYCINT. When the interface **70** receives the read command, the parallel/serial conversion circuit **72** converts the detected data ADQ held in the register **67** into serial data, and outputs the serial data as a data output signal SDO to the microcontroller **80**. The frequency of the signal SYCINT is f**2**<f**1**, and the interface **70** outputs the detected data ADQ at an output sampling rate corresponding to the frequency f**2**. The detected data ADQ output at the frequency f**2** corresponds to the detected data SD**1** and SD**2** in FIGS. **4** and **7**.

[0080] FIG. **9** shows an exemplary configuration of an angular velocity sensor device as an example of the sensor device. An angular velocity sensor device **30** includes a vibrator **56**, a drive circuit **58**, the detection circuit **60**, the processing circuit **66**, and the interface **70**.

[0081] The drive circuit **58** may include an amplification circuit that receives a feedback signal DG input from the vibrator **56** and amplifies the signal, an AGC circuit that performs automatic gain control, an output circuit that outputs a drive signal DS to the vibrator **56**, or other circuits. For example, the AGC circuit variably and automatically adjusts the gain in such a way that the amplitude of the feedback signal DG from the vibrator **56** becomes constant. The output circuit outputs the drive signal DS, which is, for example, a rectangular wave, to the vibrator **56**. The detection circuit **60** may include an amplification circuit, a synchronous wave detecting circuit, an A/D conversion circuit, and other circuits. The amplification circuit receives detection signals S**1** and S**2** input from the vibrator **56** and performs charge-voltage conversion and signal amplification on the detection signals S**1** and S**2**, which are differential signals. The synchronous wave detection circuit performs synchronous detection for extracting a desired wave by using a synchronization signal from the drive circuit **58**. The A/D conversion circuit converts the analog detection signal after the synchronous wave detection into digital detected data and outputs the detected data to the processing circuit **66**. The processing circuit **66** performs various types of processing on the detected data, such as zero point correction, sensitivity adjustment, filtering, or temperature correction, and outputs the processed detected data to the interface **70**.

[0082] In FIG. **9**, a vibrator having a double-T structure is used as the vibrator **56**. Note that a tuning-fork-type, H-type, or any other suitable vibrator may be used as the vibrator **56**. The vibrator **56** includes drive arms **38**A, **38**B, **38**C, and **38**D, detection arms **39**A and **39**B, a base **31**, and linkage arms **32**A and **32**B. The detection arms **39**A and **39**B extend in a +y-axis direction and a-y-axis direction from the base **31**, which has a rectangular shape. The linkage arms **32**A and **32**B extend in a +x-axis direction and a-x-axis direction from the base **31**. The drive arms **38**A and **38**B extend in the +y-axis direction and the −y-axis direction from a front end portion of the linkage arm **32**A, and the drive arms **38**C and **38**D extend in the +y-axis direction and the −y-axis direction from a front end portion of the linkage arm **32**B. The drive arms **38**A, **38**B, **38**C, and **38**D and the detection arms **39**A and **39**B each have a frequency adjusting weight provided at the front end thereof. Assuming that the thickness direction of the vibrator **56** extends along a Z-axis, the vibrator **56** detects angular velocity around the Z-axis.

[0083] Drive electrodes **33** are formed at the upper and lower surfaces of each of the drive arms **38**A and **38**B, and drive electrodes **34** are formed at the right and left side surfaces of each of the drive arms **38**A and **38**B. The drive electrodes **34** are formed at the upper and lower surfaces of each of the drive arms **38**C, **38**D, and the drive electrodes **33** are formed at the right and left side surfaces of each of the drive arms **38**C, **38**D. The drive signal DS from the drive circuit **58** is supplied to the drive electrodes **33**, and the feedback signal DG from each of the drive electrodes **34** is input to the drive circuit **58**. Detection electrodes **35** are formed at the upper and lower surfaces of the detection arm **39**A, and ground electrodes **37** are formed at the right and left side surfaces of the detection arm **39**A. Detection electrodes **36** are formed at the upper and lower surfaces of the detection arm **39**B, and the ground electrodes **37** are formed at the right and left side surfaces of the detection arm **39**B. The detection signals S**1** and S**2** from the detection electrodes **35** and **36** are then input to the detection circuit **60**.

[0084] The operation of the angular velocity sensor device **30** will next be described. When the drive circuit **58** applies the drive signal DS to the drive electrodes **33**, the drive arms **38**A, **38**B, **38**C, and **38**D perform flexural vibration as indicated by arrows C**1**. The drive arms each repeat, for example, a vibration mode indicated by the solid arrows and a vibration mode indicated by the dotted arrows at a predetermined frequency. In this state, when angular velocity around the Z-axis, which serves as an axis of rotation, acts on the vibrator **56**, the drive arms **38**A, **38**B, **38**C, and **38**D vibrate as indicated by arrows C**2** due to a Coriolis force. The vibration indicated by the arrows C**2** is transmitted to the base **31** via the linkage arms **32**A and **32**B, and the detection arms **39**A and **39**B perform the flexural vibration in the direction indicated by arrows C**3**. A charge signal generated due to the piezoelectric effect caused by the flexural vibration of the detection arms **39**A and **39**B is input as the detection signals S**1** and S**2** to the detection circuit **60**, so that the angular velocity around the Z-axis is detected.

4. Measurement System

[0085] FIG. **10** shows an exemplary configuration of a measurement system **200** according to the present embodiment. The measurement system **200** includes the sensor module **10**, and the host device **210** electrically coupled to the sensor module **10**. The measurement system **200** may further include a GPS receiving section **220**, an antenna **222**

for GPS reception, and an oscillator **230**. In FIG. **10**, the sensor module **10** is used as a six-axis inertia measuring unit (IMU).

[0086] The host device **210** can be realized by any of various processors such as an MPU or a CPU. Note that the host device **210** may be realized by an ASIC-type integrated circuit unit. The host device **210** includes a digital signal processor (DSP) **212**, which performs digital signal processing, and a clock signal generating circuit **213**, which generates a clock signal.

[0087] The GPS receiving section **220** receives signals from GPS satellites via the antenna **222**. That is, satellite signals carrying position information are received as a GPS carrier wave. The GPS receiving section **220** is a GPS receiver, and can be realized by an integrated circuit unit including a GPS receiving circuit. The host device **210** detects, based on the signals received by the GPS receiving section **220**, GPS positioning data representing the position, the velocity, and the direction of an object under measurement, such as a moving object. The position of the object under measurement is the latitude, the longitude, the altitude, or the like. The GPS positioning data further contains status data indicating a reception state, a reception time, or the like. The host device **210** receives acceleration data and angular velocity data from the sensor module **10**, and performs inertial navigation operation on the data to determine inertial navigation positioning data. The inertial navigation positioning data includes acceleration data and posture data of the object under measurement. The host device **210** then calculates the position, the posture, or both the position and the posture of the object under measurement based on the determined inertial navigation positioning data and the GPS positioning data. When the object under measurement is a moving object such as an automobile, the position on the ground where the moving object is traveling is calculated. The calculation of the position and the like of the object under measurement can be realized by Karman filtering using the DSP **212**.

[0088] The oscillator **230** generates an oscillation clock signal by using a vibrator such as a quartz crystal vibrator. The oscillator **230** is, for example, a temperature compensated oscillator (TCXO). An oven-controlled oscillator (OCXO) including an oven may, for example, instead be used as the oscillator **230**. The clock signal generating circuit **213** generates various clock signals to be used in the host device **210** based on the oscillation clock signal from the oscillator **230**. In this case, the clock signal generating circuit **213** generates a clock signal based on the time reference signal, which is a signal acquired from the satellite positioning system such as the GPS. The clock signal generating circuit **213** generates, for example, the external synchronization signal EXSYC as one of the clock signals.

[0089] The host device **210** can acquire accurate absolute time information based on time information contained in the satellite signals received by the GPS receiving section **220**. The time information is information such as a year, month, day, hour, minute, and second. The GPS receiving section **220** then outputs the PPS signal, which carries a pulse generated every second, as the time reference signal. The clock signal generating circuit **213** is configured with a PLL circuit that operates based on the oscillation clock signal from the oscillator **230**, and the PPS signal is input as a clock synchronization reference signal to the PLL circuit. The PLL circuit then generates a clock signal synchronized with the PPS signal, which is the time reference signal. The host device **210** thus outputs the external synchronization signal EXSYC synchronized with the time reference signal to the sensor module **10**. Note that the PPS signal from the GPS receiving section **220** may be input as the external synchronization signal EXSYC to the sensor module **10**. The PPS signal does not necessarily provide one pulse per second, and may, for example, provide ten pulses per second, as shown in FIG. **2** and other figures.

5. Electronic Instrument

[0090] FIG. **11** shows an exemplary configuration of an electronic instrument **300** according to the present embodiment. The electronic instrument **300** includes the sensor module **10** according to the present embodiment and a processing section **320**, which carries out processes based on a signal output from the sensor module **10**. The electronic instrument **300** may further include a communication section **310**, an operation section **330**, a display section **340**, a storage section **350**, and an antenna **312**.

[0091] The communication section **310** is, for example, a wireless circuit, receives data from an external apparatus, and transmits data to the external apparatus via the antenna **312**. The processing section **320** controls the electronic instrument **300**, digitally processes the data transmitted and received via the communication section **310** in various manners, and carries out other processes. The processing section **320** further carries out processes based on the signal output from the sensor module **10**. Specifically, the processing section **320** performs signal processing such as correction or filtering on the signal output from the sensor module **10**, such as the detected data, or performs various types of control on the electronic instrument **300** based on the output signal. The function of the processing section **320** can be realized, for example, by a processor such as an MPU or a CPU. The operation section **330** is used by a user to perform an input operation, and can be realized by operation buttons or a touch panel display. The display section **340** displays various pieces of information and can be realized, for example, by a display using a liquid crystal or organic EL material. The storage section **350** stores data, and the function thereof can be realized by a semiconductor memory such as a RAM or a ROM.

[0092] Note that the electronic instrument **300** according to the present embodiment is applicable, for example, to a video-related instrument such as a digital camera or a video camcorder, an in-vehicle instrument, a wearable instrument such as a head mounted display or a timepiece-related instrument, an inkjet-type discharge apparatus, a robot, a personal computer, a portable information terminal, a printing apparatus, or a projection apparatus. The in-vehicle instrument is, for example, a car navigator or an automatic driving instrument. The timepiece-related instrument is, for example, a timepiece or a smartwatch. The inkjet-type discharge apparatus is, for example, an inkjet printer. The mobile information terminal is, for example, a smartphone, a mobile phone, a mobile game console, a laptop PC, or a tablet terminal. The electronic instrument **300** according to the present embodiment is further applicable to an electronic organizer, an electronic dictionary, an electronic calculator, a word processor, a workstation, a videophone, a security television monitor, electronic binoculars, a POS terminal, a medical instrument, a fish-finder, a measurement instrument, an instrument for mobile terminal base station, meters and gauges, a flight simulator, and a network server. The medical instrument is, for example, an electronic thermometer, a blood pressure monitor, a blood glucose meter, an electrocardiogram measuring apparatus, an ultrasonic diagnostic apparatus, or an electronic endoscope. The meters and gauges are for a vehicle, an aircraft, a ship, and the like.

[0093] While the present embodiment has been described above in detail, a person skilled in the art can readily understand that many changes can be made thereto without substantially departing from the novel items and advantages of the present disclosure. All such variations therefore fall within the scope of the present disclosure. For example, a term described at least once together with a different term having a broader meaning or the same meaning in the specification or the drawings can be replaced with the different term at any place in the specification or the drawings. All combinations of the present embodiment and the variations thereof also fall within the scope of the present disclosure. The configurations and operations of the sensor element, the detection circuit, the interface, the sensor device, the digital interface, the processing circuit, the signal processing circuit, the interrupt controller, the host interface, the microcontroller, the host device, the measurement system, the electronic instrument, and the like are not limited to those described in the present embodiment, and can be changed in various manners.

What is claimed is:

1. A sensor module comprising:

a first sensor device including a first sensor element, a first detection circuit configured to receive a signal input from the first sensor element and perform detection, and a first interface configured to output first detected data from the first detection circuit;

a second sensor device including a second sensor element, a second detection circuit configured to receive a signal input from the second sensor element and perform detection, and a second interface configured to output second detected data from the second detection circuit; and

a microcontroller including a synchronization terminal to which an external synchronization signal is input and configured to receive the first detected data input from the first sensor device and the second detected data input from the second sensor device,

wherein the microcontroller is configured to determine first integrated data by integrating the first detected data up to a synchronization timing of the external synchronization signal, determine second integrated data by integrating the second detected data up to the synchronization timing, and output the first integrated data and the second integrated data to a host device at the synchronization timing.

2. The sensor module according to claim **1**, wherein

the microcontroller is configured to determine the first integrated data by integrating the first detected data for a period between the synchronization timings of the external synchronization signal, and determine the second integrated data by integrating the second detected data for the period between the synchronization timings.

3. The sensor module according to claim **2**, wherein

the microcontroller is configured to reset the first integrated data and the second integrated data whenever the synchronization timing is reached.

4. The sensor module according to claim **2**, wherein

the microcontroller is configured to output a signal that notifies completion of the integration to an external apparatus when the integration between the synchronization timings is completed.

5. The sensor module according to claim **1**, wherein

the microcontroller is configured to determine the first integrated data by integrating the first detected data for a period between output timings including the synchronization timing of the external synchronization signal and a timing after a predetermined period from the synchronization timing, and determine the second integrated data by integrating the second detected data for the period between the output timings.

6. The sensor module according to claim **5**, further comprising

a register configured to store set information on the predetermined time,

wherein the microcontroller is configured to set a timing after the predetermined time based on the set information.

7. The sensor module according to claim **5**, wherein

the microcontroller is configured to reset the first integrated data and the second integrated data whenever the output timing is reached.

8. The sensor module according to claim **5**, wherein

the microcontroller is configured to output a signal that notifies completion of the integration to an external apparatus when the integration between the output timings is completed.

9. The sensor module according to claim **1**, wherein

the external synchronization signal is a time reference signal acquired from a satellite positioning system.

10. A measurement system comprising:

the sensor module according to claim **1**; and

the host device electrically coupled to the sensor module.

11. The measurement system according to claim **10**, wherein

the host device is configured to determine at least one of a position and a posture of a moving object based on positioning information from a satellite positioning system and a signal output from the sensor module.

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